LinkLens

Watch LinkedIn Through Your Own Lens

Product Requirements Document

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Status	Priority	Timeline	Target Users
In Development	High	8-10 Weeks	Technical Professionals

1. Problem Statement

LinkedIn has 1.3 billion professionals and serves content through an AI-powered feed. But for technical professionals like Agentic AI developers, the feed is dominated by:

- Motivational coaches and hustle culture content with zero technical value
- Anxiety-inducing job market posts that increase stress without adding signal
- Mass hiring broadcasts irrelevant to your specific role and goals
- Recycled thought leadership with no original insight
- Engagement bait designed to game the algorithm, not inform you

The result: you spend 20-30 minutes scrolling LinkedIn daily and extract almost zero signal relevant to your actual work in Agentic AI.

Note: LinkedIn rebuilt its feed algorithm in March 2026 using LLMs and GPU-powered ranking. While this improves semantic relevance, approximately 40% of the average feed is still sponsored content, and the algorithm optimizes for platform engagement — not YOUR career goals.

The core problem: LinkedIn's algorithm optimizes for engagement, not relevance to YOUR career goals.

2. Target User

Primary Persona

Senior technical professionals (Agentic AI developers, ML engineers, SWEs) who:

- Check LinkedIn daily for career intelligence and industry updates
- Are overwhelmed by the noise-to-signal ratio in their feed
- Want to stay current on Agentic AI / ML without the noise

- Are actively job searching or building a personal brand
- Value deep technical content over motivational fluff

User Persona

Name: Shravya — AI Team Lead, USC MS CS, actively job searching, building ShravyaVerse brand. Checks LinkedIn daily, drowns in coaching posts and anxiety content, wants only Agentic AI and ML engineering signal.

3. Proposed Solution

A personal multi-agent intelligence system that sits between you and your LinkedIn feed:

- Reads posts via a browser extension (safe, uses local cookies)
- Classifies each post by topic, signal score, relevance score, and anxiety score
- Hides noise automatically — noisy posts fade out
- Surfaces only what matters — important posts get highlighted
- Generates a daily AI brief with only the signal that matters
- Learns your preferences over time through a memory agent

4. Market Analysis

Existing Competitors

Tool	What It Does	Gap vs. Our Solution
LiSeller	Auto-comments on posts using AI	Outreach focused, not feed curation
Linkmate	Automates engagement to train LinkedIn's algorithm	Trains LinkedIn's algorithm, not YOUR personal filter
Expandi	Outreach automation sequences	Sales/marketing focused, not personal intelligence
Meet Alfred	Multi-channel outreach automation	Not a feed intelligence or noise filtering tool
Engage AI	AI comment generator for posts	Engagement tool, not a signal filter
Supergrow	LinkedIn content creation tool	Creator-focused (output), not consumer-focused (input)
PhantomBuster	LinkedIn scraping + automation	High account ban risk (340% increase in 2026)
LinkedIn Premium	InMail + recruiter search	No feed intelligence or noise filtering whatsoever

Key Market Insight

All existing tools focus on OUTBOUND — posting, commenting, outreach. Nobody is solving the inbound problem: filtering what YOU consume. This is the gap LinkLens fills.

5. SWOT Analysis

STRENGTHS ✅	WEAKNESSES ⚠️
• No direct competitor solving personal feed intelligence for technical professionals • You are your own user — authentic problem, authentic demo • Multi-agent architecture naturally showcases Agentic AI skills • Browser extension model is safer than LinkedIn API scraping • Personalization memory creates switching costs — gets better over time • Low infrastructure cost — runs locally, no heavy cloud spend	• LinkedIn TOS restricts scraping — need careful browser extension implementation • Browser extension requires installation — user adoption friction • Cold start problem — needs user interaction data before personalization kicks in • LinkedIn's new LLM-powered algorithm (March 2026) already tries to improve relevance • Solo developer building it — speed constraint
OPPORTUNITIES 🚀	**THREATS 🔴**
• LinkedIn still shows ~40% sponsored content — noise problem is not solved • Growing demand for 'digital focus' tools among knowledge workers • Could expand to Twitter/X, newsletters, and other feeds • SaaS monetization potential: $8-15/month for technical professionals • Could grow into a full Career Intelligence Platform • Open source version builds community and drives adoption	• LinkedIn actively detects and bans automation (340% increase in restrictions 2026) • LinkedIn could build native feed filtering — making this redundant • Low switching cost — users could just unfollow noisy accounts manually • Microsoft (LinkedIn owner) could restrict browser extension access

7. Core UX — How Noise Is Hidden

The Passive Observation Model

The most important architectural decision: the agent NEVER interacts with LinkedIn. It only watches what YOU see as you scroll normally. From LinkedIn's perspective, a human is scrolling — no bots, no API calls, no automation detected.

What You See Without the Agent

Post in Your Feed	Your Experience

Coaching post — '5 habits of successful people'	Full size, you waste time reading it
Agentic AI post — 'LangGraph update'	Full size, you might miss it
Anxiety post — 'AI will take your job'	Full size, causes stress
ML Engineering — 'RAG best practices'	Full size, buried in noise
Mass hiring broadcast	Full size, irrelevant

What You See WITH the Agent

Post in Your Feed	Your Experience
Coaching post	Fades out instantly, collapses to 1 line — your eyes skip it
Agentic AI post ✨	Highlighted, full size — you stop and read it
Anxiety post	Fades out, collapses — you never stress about it
ML Engineering post ✨	Highlighted, full size — you engage with it
Mass hiring broadcast	Fades out, collapses — invisible to you

How It Works Technically — MutationObserver

The browser extension uses MutationObserver — a standard, legitimate browser API used by thousands of extensions including ad blockers. Here's the flow:

- You open LinkedIn and scroll normally as a human
- MutationObserver watches for new posts appearing in the DOM as you scroll
- When a new post appears, the extension silently captures its text
- Text is sent to your LOCAL FastAPI backend — not to any cloud server
- JudgeAgent scores it in milliseconds — topic, signal score, anxiety score
- Extension overlays the result: noisy posts fade/collapse, signal posts highlight
- LinkedIn never knows — from its servers' perspective, a human is just scrolling

This is identical to how ad blockers work — they read the DOM and hide elements. Ad blockers are completely legal and used by hundreds of millions of people. Your tool does the same thing for professional noise.

Why This Is NOT a Bot

Behaviour	Traditional Bot ❌	LinkLens ✅
Sends requests to LinkedIn servers	Yes — detectable	No — never

Automates clicks or scrolling	Yes — detectable	No — you scroll
Scrapes data at scale	Yes — violates TOS	No — reads your screen only
Runs in the cloud	Yes — shared IPs flagged	No — runs locally on your machine
Posts or comments automatically	Often — high ban risk	Never — read only
Sells or redistributes data	Sometimes	Never — open source, personal use

8. Agent Architecture

The 5-Step Flow

Step	Agent	What It Does
1 — Observe	FeedObserver (ScoutAgent)	Browser extension reads visible LinkedIn posts, saved posts, email digests
2 — Understand	TopicClassifier (JudgeAgent)	LLM tags each post: topic, signal_score (0-1), relevance_score (0-1), anxiety_score (0-1)
3 — Decide	NoiseFilter (RuleEngine)	IF coaching → HIDE | IF agentic_ai → HIGHLIGHT | IF mass hiring → LOW_PRIORITY
4 — Act	ClusterEngine + InsightGenerator	Hides noisy posts, clusters duplicates, generates daily AI brief
5 — Remember	MemoryAgent (PreferenceMemory)	Learns from user behavior — unhide = increase weight, ignore = decrease

Multi-Agent System Design

The production architecture uses 4 cooperating autonomous agents:

- ScoutAgent — continuously scans and collects LinkedIn post content
- JudgeAgent — classifies signal vs noise using LLM reasoning
- SynthesizerAgent — generates daily brief with only high-signal content
- MemoryAgent — tracks user preferences and trending topics over time

9. MVP Scope (v1)

In Scope

- Browser extension that reads visible LinkedIn post text
- Local FastAPI backend that receives and scores posts

- JudgeAgent classifies posts with topic tag and signal score
- Noisy posts fade out visually in the browser
- Important posts get a highlighted badge
- Basic preference memory (like/dislike on posts)

Out of Scope for v1

- Autonomous posting or commenting on LinkedIn
- Email digest integration
- Mobile app
- Multi-platform support (Twitter/X, newsletters)
- SaaS infrastructure / payments

Success Metrics

- Time spent on LinkedIn reduced by 30%+ per session
- Signal-to-noise ratio improves — user sees 80%+ relevant content
- Daily brief generated with 5-10 high-signal posts
- Memory personalization kicks in after 50+ rated posts

10. Tech Stack

Component	Technology	Reason
Browser Extension	JavaScript / Chrome Extension API	Reads posts safely using local cookies — lower ban risk
Backend API	FastAPI (Python)	Fast, async, easy to deploy locally
LLM Integration	Anthropic Claude API	Best-in-class classification and reasoning
Agent Framework	LangChain / LangGraph	Multi-agent orchestration and memory
Memory Store	ChromaDB (vector) + SQLite	Stores embeddings for preference memory
Embeddings	HuggingFace sentence-transformers	Fast local embeddings for clustering
Frontend UI	React (extension popup)	Clean settings and daily brief display

11. Roadmap

Phase	Timeline	Deliverables
v1 — MVP	Weeks 1-3	Browser extension + JudgeAgent + NoiseFilter. Posts scored and filtered locally.
v2 — Memory	Weeks 4-5	MemoryAgent + PreferenceMemory. Personalization based on user feedback.
v3 — Daily Brief	Weeks 6-7	SynthesizerAgent generates daily AI brief. ClusterEngine deduplicates topics.
v4 — Multi-Agent	Weeks 8-9	Full ScoutAgent + JudgeAgent + SynthesizerAgent + MemoryAgent pipeline.
v5 — SaaS	Week 10+	Cloud deployment, user accounts, $8-15/month pricing tier.

12. Why This Project Matters

As a Portfolio Project

- Demonstrates real multi-agent system design — not a toy
- Shows human-aligned AI filtering and cognitive load reduction
- Browser extension + local agent = production-grade architecture
- You are your own user — authentic demo in every interview
- Directly aligned with ThoughtAgent brand in ShravyaVerse

As a Product

- No direct competitor solving this specific problem
- Real monetization path: $8-15/month SaaS for technical professionals
- Open source version drives community adoption and brand building
- Foundation for a full Career Intelligence Platform